UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 37605

LH EXOPLANETS

SEMESTER 2 EXAMINATIONS 2022/23

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

A list of useful formulae is provided at the end of the paper.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

ANY CALCULATOR

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. (a) Describe the physical processes and steps behind the creation of a gas-giant planet, starting from the original nebula. Cite relevant equations from the formula sheet where appropriate. **[7]**

   (b) Derive the scale-height of a protoplanetary disc, starting from the equation:
   $\rho(z) = \mu\, m_{p^+}\, n(0)\, e^{-E(z)/kT(R)}$,
   explaining the meaning of each variable in the equation.
   State clearly any assumption you make. **[10]**

   (c) Create a scaling relation for the scale-height of a gas disc, as a function of orbital distance in AU, calibrating it to a Sun-like star, for orbital periods of $1\ \mathrm{year}$. Show how to use your scaling relation to calculate the scale-height of a planet situated $200\ \mathrm{AU}$ from the star. **[6]**

   (d) Estimate the Toomre criterion for a planet situated $200\ \mathrm{AU}$ from its Sun-like star. Assume $\Sigma = 10\ \mathrm{g\,cm^{-2}}$ (of dust plus gas). What can you conclude about how this planet formed? **[7]**

2. A planet with mass $m_{\rm p} = 10\ {\rm M}_\oplus$, radius $r_{\rm p} = 2.5\ {\rm R}_\oplus$, on an eccentric orbit with $e = 0.98$, $\omega = 270°$, has an orbital period $T = 50\ {\rm days}$. The planet orbits a $5\ {\rm Gyr}$-old, Sun-like star with $v \sin i_\star = 5\ {\rm km\,s^{-1}}$. The orbital inclination $i_{\rm p} = 90°$.

   (a) Use the Roche radius equation to create a scaling relation, in units of solar radii, calibrated on a Jupiter-like planet orbiting a Sun-like star. Then show how to use your relation to find out whether the planet under consideration is in danger of being destroyed. **[6]**

   (b) Starting from $\tau_{\rm circ} = E/\dot{E}$, where E denote energy, use equations in the formula sheet to show that the tidal circularisation timescale can be written $\tau_{\rm circ} = \frac{5}{6\pi}\frac{QT}{k_2}\left(\frac{a_{\rm peri}}{r_{\rm p}}\right)^5 \frac{1-e}{1+e}$.
   Justifying the value of k_2 you use, calculate $\tau_{\rm circ}$ for the planet, giving your answer in ${\rm Gyr}$. What can you conclude from this value, what will likely happen to the planet?
   State clearly any assumptions you make. Assume the tidal $Q = 10^4$. **[12]**

   (c) A telescope observes the Rossiter-McLaughlin effect caused by this planet. The telescope collects data every $300\ {\rm s}$, with a precision of $20\ {\rm cm\,s^{-1}}$. Calculate the signal-to-noise ratio expected after observing a single event. **[6]**

   (d) Selecting only the Helium line to compute radial-velocities, the semi-amplitude of the Rossiter-McLaughlin effect is measured to be 50% larger if using the whole spectrum. Explain in a few sentences what could be happening. Support your argument using the Hill sphere equation. **[6]**

3. A star hosts a transiting planet with orbital period $T = 33.33\ \mathrm{days}$. It is being observed with photometry and with Doppler spectroscopy. The transit has a depth of $250\ \mathrm{ppm}$ and a duration of $5.75\ \mathrm{hours}$; the impact parameter is zero. The planet is on a circular orbit, with a semi-amplitude $K_\star = 1\ \mathrm{m\,s^{-1}}$. The star has p-modes where $\nu_{\mathrm{max}} = 3.10\ \mathrm{mHz}$ and $\Delta\nu = 137.6\ \mu\mathrm{Hz}$. The system's visual magnitude is $V = 6$, and its parallax is $\pi = 42.3\ \mathrm{mas}$.

   [Solar values are $\nu_{\mathrm{max},\odot} = 3.05\ \mathrm{mHz}$ and $\Delta\nu_\odot = 134.9\ \mu\mathrm{Hz}$]

   (a) Rearrange the equation for a planet surface gravity, to express it as a function of transit depth and duration. Use your expression to calculate the planet's surface gravity. **[5]**

   (b) Calculate the mass, and the radius of the planet, in Earth units. Explain clear your steps and each assumption you make. What sort of planet is this, and why are they interesting to study? **[10]**

   (c) This system contains a second star in addition to the planet-hosting star. Although hundreds of AU distant from one another, seen from Earth the pair is not visually nor spectrally resolved. First show how it is possible to deduce the presence of this other star. Secondly, measure its brightness relative to the planet-hosting star, and quantitatively describe what consequences this will have for the planet, and for your interpretation of its results. **[10]**

   (d) Most exoplanet parameters are determined relative to the stellar parameters. How can absolute masses and radii be determined for an exoplanet? **[5]**

Formula Sheet

1 Planet formation

1.1 Angular momentum: $L = m\,v\,a$

1.2 Keplerian velocity: $v_{\rm Kep} = \sqrt{\frac{G\,M_\star}{a}}$

1.3 Orbital Angular momentum: $L = m_{\rm p}\sqrt{GM_\star a}$

1.4 Virial Theorem: $2K + U = 0$

1.5 Gravitational potential (uniform sphere): $U_{\rm grav} = -\frac{3}{5}\,\frac{GM^2}{R}$

1.6 Kinetic energy of a gas particle: $K_{\rm gas} = \frac{3}{2}\,k\,T$

1.7 Jeans length: $R_{\rm jeans} = \sqrt{\frac{15\,k}{4\,\pi\,G}\,\frac{T}{\rho\,\mu\,m_{\rm p+}}}$

1.8 Jeans mass: $M_{\rm jeans} = \left(\frac{3}{4\,\pi\,\rho}\right)^{1/2}\left(\frac{5\,k\,T}{G\,\mu\,m_{\rm p+}}\right)^{3/2}$

1.9 Mass density: $\rho(z) = \mu\,m_{p^+}\,n(0)\,e^{-E(z)/kT(R)}$

1.10 Gravitational energy perpendicular to disc plane: $E(z) = \frac{G\,\mu\,m_{p^+}\,M_\star}{2R^3}\,z^2$

1.11 Protoplanetary disc scale height: $H = \sqrt{\frac{kT(a)a^3}{GM_\star\mu m_{\rm p+}}}$

1.12 Luminosity law: $L = 4\pi\,R^2\,\sigma\,T^4$

1.13 Equilibrium Temperature: $T_{\rm eq} = T_\star(1-A)^{1/4}\sqrt{\frac{R_\star}{2a}}$

1.14 Angular velocity: $\Omega = \frac{2\pi}{P} = \sqrt{\frac{G\,M_\star}{a^3}}$

1.15 The Hill sphere radius: $r_{\rm H} = a\sqrt[3]{\frac{m_{\rm p}}{3M_\star}}$

1.16 Escape velocity: $v_{\rm esc} = \sqrt{\frac{2Gm_{\rm p}}{r_{\rm p}}}$

1.17 Surface gravity: $g = \frac{Gm_{\rm p}}{r_{\rm p}^2}$

1.18 Mass accretion: $\frac{dm}{dt} = \pi\,r^2\,v\,\rho = \sigma\,v\,\rho$

1.19 Energy conservation: $\frac{1}{2}\,m'\,v_{\rm max}^2 - \frac{G\,m\,m'}{\Delta r} = \frac{1}{2}\,m'\,v^2$

1.20 Angular momentum conservation: $v_{\rm max}^2 = \frac{v^2\, b^2}{\Delta r^2}$

1.21 Impact Parameter: $b^2 = r^2 \left(1 + \frac{v_{\rm esc}^2}{v^2}\right) = r^2\,(1+\Theta)$

1.22 Safronov Number: $\Theta = \frac{v_{\rm esc}^2}{v^2}$

1.23 Mass accretion with gravitational focusing: $\frac{dm}{dt} = \pi r^2 v \rho\,(1+\Theta)$

1.24 Runaway core accretion: $\dot{m} \propto m^{4/3}$

1.25 Sound speed (disc): $c_{\rm s} = \frac{H}{R} v_{\rm Kep}$

1.26 Observed mass accretion: $\frac{dm}{dt} = \frac{1}{2}\,\Sigma\,\Omega\,\pi\,r^2\,(1+\Theta)$

1.27 Isolation mass is: $m_{\rm iso} = 8\frac{\pi^{3/2}}{\sqrt{3}} a^3 M_\star^{-1/2} \Sigma^{3/2}$

1.28 Gas Accretion: $m \geq \sqrt{\frac{3}{32\,\pi}\,\frac{M_\star^3}{\rho\,a^3}}\,\left(\frac{H}{a}\right)^3$

1.29 Toomre criterion: $Q = \frac{c_{\rm s}\,\Omega}{\pi\,G\,\Sigma} < 1$

1.30 Disc Collapse conditions: $\frac{m_{\rm d}}{M_\star} > \frac{h}{R}$

2 Two-body problem

2.1 Newton's law: $\boldsymbol{F}_1 = m_1\,\ddot{\boldsymbol{r}}_1 = +\frac{G m_1 m_2}{r^3}\boldsymbol{r}$

2.2 Newton's law: $\boldsymbol{F}_2 = m_2\,\ddot{\boldsymbol{r}}_2 = -\frac{G m_1 m_2}{r^3}\boldsymbol{r}$

2.3 Equation of Motion: $\ddot{\mathbf{r}} + \mu\,\frac{\dot{\mathbf{r}}}{r^3} = 0$

2.4 Mass Scale: $\mu = G\,(m_1 + m_2)$

2.5 Planar motion: $\mathbf{r} \times \dot{\mathbf{r}} = \mathbf{h}$

2.6 Polar coordinates: $\mathbf{r} = r\,\hat{\mathbf{r}}$

2.7 Polar coordinates: $\dot{\mathbf{r}} = \dot{r}\,\hat{\mathbf{r}} + r\,\dot{\theta}\,\hat{\theta}$

2.8 Polar coordinates: $\ddot{\mathbf{r}} = \left(\ddot{r} - r\,\dot{\theta}^2\right)\hat{\mathbf{r}} + \left(\frac{1}{r}\,\frac{d}{dt}\left(r^2\dot{\theta}\right)\right)\hat{\theta}$

2.9 Angular momentum: $\mathbf{h} = r^2\,\dot{\theta}\,\hat{\mathbf{z}}$

2.10 Conic's equation: $r = \frac{h^2/\mu}{1+e\cos(\theta-\varpi)}$

2.11 Specific angular momentum: $h = \sqrt{\mu a(1-e^2)}$

2.12 Distance to the star: $r = \frac{a\,(1-e^2)}{1+e\,\cos f}$

2.13 Kepler's second law: $\dot{A} = \frac{1}{2}\sqrt{\mu a(1-e^2)}$

2.14 Kepler's third law: $P^2 = \frac{4\pi^2}{\mu} a^3$

2.15 Periastron velocity: $v_{\rm peri}^2 = \frac{G\,M_\star}{a}\,\frac{(1+e)}{(1-e)}$

2.16 Vis-Viva equation: $v_{\rm pl}^2 = G\,M_\star\,\left(\frac{2}{r} - \frac{1}{a}\right)$

2.17 Area of an ellipse: $A = \pi a b = \pi a^2\sqrt{1-e^2}$

2.18 Orbital angles relations: $\varpi = \Omega + \omega; \theta = f + \varpi$

3 Indirect detection methods

3.1 Doppler effect: $\lambda_{\rm obs} = \lambda_0\,\frac{1+\frac{v}{c}}{\left(1-\frac{v^2}{c^2}\right)^{1/2}}$

3.2 Radial velocity: $V_{\rm rad} = v\sin i = \frac{\Delta\lambda}{\lambda_0}\,c$

3.3 R: $\mathbf{R} = \frac{m_1\,\mathbf{r}_1+m_2\,\mathbf{r}_2}{m_1+m_2}$

3.4 $\ddot{R}$: $\ddot{\mathbf{R}} = \frac{m_1\,\ddot{\mathbf{r}}_1+m_2\,\ddot{\mathbf{r}}_2}{m_1+m_2} = 0$

3.5 Reflex motion: $R_2 = \frac{m_1}{M_{\rm tot}}\,a = a_2$

3.6 Reflex motion: $R_1 = \frac{m_2}{M_{\rm tot}}\,a = a_1$

3.7 Radial velocity: $V_{\rm rad} = \dot{z} = \sin i\left[\dot{r}\,\sin(f+\omega) + r\,\dot{f}\,\cos(f+\omega)\right]$

3.8 $\dot{r}$: $\dot{r} = \frac{r\,\dot{f}\,e\,\sin f}{1+e\,\cos f}$

3.9 Radial velocity equation: $V_{\rm rad} = \frac{2\pi a}{P}\frac{\sin i}{\sqrt{1-e^2}}[\cos(f+\omega) + e\cos\omega] + \gamma$

3.10 Semi-amplitude: $K = \frac{2\,\pi\,a}{P}\frac{\sin i}{\sqrt{1-e^2}}$

3.11 Motion caused by a planet: $K_\star = \frac{m_{\rm p}}{M_\star+m_{\rm p}}\frac{2\pi}{P}\frac{a\sin i}{\sqrt{1-e^2}}$

3.12 Motion caused by a star: $K_{\rm p} = \frac{M_\star}{M_\star+m_{\rm p}}\frac{2\pi}{P}\frac{a\sin i}{\sqrt{1-e^2}}$

3.13 Scaling relation: $K_1 = 28.4\,{\rm m\,s^{-1}}\left(\frac{P}{1\ {\rm year}}\right)^{-1/3}\left(\frac{m_{\rm p}\,\sin i}{{\rm M_{jup}}}\right)\left(\frac{M_\star}{{\rm M_\odot}}\right)^{-2/3}\frac{1}{\sqrt{1-e^2}}$

3.14 Planet's mass function: $f(m_{\rm p}) = \frac{m_{\rm p}^3 \sin^3 i}{(M_\star + m_{\rm p})^2} = (1-e^2)^{3/2} \frac{P K_\star^3}{2\pi G}$

3.15 Transit probability: $\mathcal{P} \sim \frac{R_\star}{a}$

3.16 Transit depth: $D = \frac{r_{\rm p}^2}{R_\star^2}$

3.17 Scaled radius: $\frac{R_\star}{a} = \frac{W}{P} \frac{\pi}{\sqrt{(1+\sqrt{D})^2 - b^2}} \frac{1+e\sin\omega}{\sqrt{1-e^2}}$

3.18 Distance from stellar centre: $\mu = \sqrt{1-(X^2+Y^2)}$

3.19 Limb darkening law: $I = 1 - u_a(1-\mu) - u_b(1-\mu)^2$

3.20 Stellar density: $\rho_\star \sim \left(\frac{a}{R_\star}\right)^3 \frac{4\pi^2}{GP^2}$

3.21 Planet gravity: $g_{\rm p} = \frac{2\pi}{P} \frac{K_\star \sqrt{1-e^2}}{\sin i} \left(\frac{a}{r_{\rm p}}\right)^2$

3.22 Eclipse Depths: $D_{\rm pri} = \frac{(F_1+F_2) - \left(\mathcal{A}_{\rm pri} F_1 + F_2\right)}{F_1+F_2}$ $\quad D_{\rm sec} = \frac{(F_1+F_2) - (\mathcal{A}_{\rm sec} F_2 + F_1)}{F_1+F_2}$

3.23 Eclipse duration ratio: $\frac{W_{\rm occ}}{W_{\rm tra}} \sim \frac{1+e\sin\omega}{1-e\sin\omega}$

3.24 Eclipse time: $\Delta t \sim \frac{T}{2}\left(1 + \frac{4}{\pi} e\cos\omega\right)$

3.25 Projected separation: $\delta(f) = \frac{1-e^2}{1+e\cos f} \sqrt{1 - \sin^2 i \sin^2(f+\omega)}$

3.26 Eclipse criterion: $\delta(f) < \frac{R_1+R_2}{a}$

3.27 Transit Ephemeris: $T_n = T_0 + n \times T$

3.28 Right Ascension: $\alpha^\star = \alpha \cos\delta$

3.29 Parallax: $\varpi = 1'' \left(\frac{1\ {\rm pc}}{d}\right)$

3.30 Astrometric displacement: $\Lambda = 1\,{\rm arcsec} \left(\frac{m_{\rm p}}{M_\star}\right) \left(\frac{a}{1\,{\rm AU}}\right) \left(\frac{d}{1\,{\rm pc}}\right)^{-1}$

3.31 Light deflection angle: $\epsilon = \frac{4GM_{\rm L}}{rc^2}$

3.32 Lens equation: $\theta_1^2 - \alpha\,\theta_1 - \theta_{\rm E}^2 = 0$

3.33 Einstein radius: $\theta_{\rm E} = \sqrt{\frac{4GM_{\rm L}}{\mu D_{\rm L} c^2}}$ with $\mu = 1 + \frac{D_{\rm L}}{D_{\rm LS}}$

3.34 Einstein radius: $r_{\rm E} = \theta_{\rm E}\, D_{\rm L}$

3.35 Einstein radius scaling relation: $r_{\rm E} = 2.2\,{\rm AU} \left(\frac{M_{\rm L}}{0.3\,{\rm M}_\odot}\right)^{1/2} \left(\frac{D_{\rm S}}{8\,{\rm kpc}}\right)^{1/2} \left(\frac{x(1-x)}{0.25}\right)^{1/2}$

3.36 Image position: $\theta_{\pm} = \pm\frac{\theta_{\rm E}}{2}\left(\sqrt{u^2+4} \pm u\right)$

3.37 Image position: $\theta_{\pm} = \pm\frac{1}{2}\alpha + \left[\left(\frac{1}{2}\alpha\right)^2 + \theta_{\rm E}^2\right]^{1/2}$

3.38 Lens magnification: $A_{\pm} = \left|\frac{\theta_{\pm}}{\alpha}\,\frac{d\theta_{\pm}}{d\alpha}\right|$

3.39 Microlensing amplitude: $A = \frac{u^2+2}{u\sqrt{u^2+4}}$

3.40 Duration of microlensing: $t_{\rm E} = \theta_{\rm E}/\nu$

4 Atmospheres

4.1 Eclipse Depth: $D(\lambda) = \frac{F_{\rm out}-F_{\rm in}}{F_{\rm out}}$

4.2 Transmission spectroscopy: $D(\lambda) = D + D'(\lambda)$

4.3 Transmission spectroscopy: $D(\lambda) = D + \frac{2r_{\rm p}HA_{\lambda}}{R_{\star}^2}$

4.4 Atmospheric scale height: $H = \frac{kT}{\mu m_{\rm p+} g}$

4.5 Thermal velocity: $v_{\rm therm} = \sqrt{\frac{3kT}{\mu m_{p+}}}$

4.6 Emission flux: $\Delta F_{\rm occ}(\lambda) = D\frac{B_{\lambda}(T_{\rm p})}{B_{\lambda}(T_{\star})} \sim D\frac{T_{\rm p}}{T_{\star}}$

4.7 Reflected flux: $\Delta F_{\rm ref}(\lambda) = A_{\lambda}\left(\frac{r_{\rm p}}{a}\right)^2$

4.8 Occultation depth: $\Delta F_{\rm occ}(\lambda) = \Delta F_{\rm em}(\lambda) + \Delta F_{\rm ref}(\lambda)$

4.9 Apparent magnitude: $m_1 - m_2 = -2.5\log_{10}\left(\frac{F_1}{F_2}\right) = -2.5\log_{10}\Delta F_{\rm occ}$

4.10 Absolute magnitude: $m_1 - M_1 = 5\log_{10} d/10$

4.11 Signal to Noise ratio: $SNR \sim \frac{A}{\sigma_i/\sqrt{N}}$

4.12 Inner edge of the habitable zone: $a_{\rm HZ,in} = a_{\oplus} \times \sqrt{\frac{1}{2}}$

4.13 Outer edge of the habitable zone: $a_{\rm HZ,out}s = a_{\oplus} \times \sqrt{2}$

5 Know thy star

5.1 Planck's law: $B_\lambda(T) = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/(\lambda kT)}-1}$

5.2 Wien's law: $\lambda_{\rm top} = \frac{\kappa}{T_{\rm eff}}$ where $\kappa = 0.0028978\,{\rm m\,K}$

5.3 Sound speed (atmosphere): $c_{\rm s}^2 = \frac{\partial P}{\partial \rho}$

5.4 Ideal gas law: $P = \frac{n\,\mathcal{R}\,T}{V}$

5.5 Sound speed (atmosphere): $c_{\rm s}^2 = \gamma\frac{\mathcal{R}\,T}{\mathcal{M}}$

5.6 Pressure (inside a star): $P \propto \frac{GM_\star^2}{R_\star^4}$

5.7 Large separation scaling relation: $\Delta\nu = \Delta\nu_\odot\left(\frac{M_\star}{M_\odot}\right)^{1/2}\left(\frac{R_\star}{R_\odot}\right)^{-3/2}$

5.8 Acoustic cut-off frequency: $\nu_{\rm ac} = \frac{c_{\rm s}}{2\,H_{\rm p}}$

5.9 Pressure scale-height: $H_{\rm p} = \frac{k\,T}{\mu\,m_{p+}\,g}$

5.10 Maximum frequency scaling relation: $\nu_{\rm max} = \nu_{{\rm max},\odot}\left(\frac{M_\star}{M_\odot}\right)\left(\frac{R_\star}{R_\odot}\right)^{-2}\left(\frac{T_{\rm eff}}{T_{{\rm eff},\odot}}\right)^{-1/2}$

5.11 Scaling relation: $V_{\rm osc} = 23.4\left(\frac{L_\star}{L_\odot}\right)\left(\frac{M_\star}{M_\odot}\right)^{-1}\,{\rm cm\,s^{-1}}$

5.12 Scaling relation: $\frac{\delta L_\star}{L_\star} = 4.7\left(\frac{L_\star}{L_\odot}\right)\left(\frac{M_\star}{M_\odot}\right)^{-1}\left(\frac{T_{\rm eff}}{T_\odot}\right)^{-2}\left(\frac{\lambda}{550\,{\rm nm}}\right)^{-1}\,{\rm ppm}$

5.13 Radial-velocity precision: $\sigma_{\rm RV} \sim \frac{\sqrt{\rm FWHM}}{C\times{\rm SNR}}$

5.14 Skumanich law: $\Omega_\star = \frac{v_\star}{R_\star} \propto t^{-1/2}$

6 Non-Keplerian effects & Migration

6.1 Type I migration timescale: $\tau_{\rm I} = \frac{1}{2}\frac{H/a}{m_{\rm p}/M_\star}\frac{M_\star}{\Sigma a^2}\Omega_{\rm p}^{-1}$

6.2 Type I migration timescale: $\tau_e \sim 2.6(H/a)^2\tau_{\rm I}$ and $\tau_i \sim 3.7(H/a)^2\tau_{\rm I}$

6.3 Type II migration timescale: $\tau_{\rm II} = \frac{a^2}{\alpha H^2\Omega}\times\max\left(1, \frac{m_{\rm p}}{4\pi\Sigma a^2}\right)$

6.4 Planet separation: $\kappa = \frac{a_2-a_1}{\Delta r_{\rm H}}$

6.5 Effective planet separation: $\kappa_{\rm eff} = \kappa - 0.5\sigma_\kappa$

6.6 Mutual Hill radii: $\Delta r_{\rm H} = \frac{a_1+a_2}{2}\left(\frac{m_{\rm p,1}+m_{\rm p,2}}{3M_\star}\right)^{1/3}$

6.7 Tidal circularisation semi-major axis: $a_f = a_p\,(1+e_i)$

6.8 Tidal acceleration: $a_{\rm tides} = \pm\frac{2GM_\star\Delta r}{r^3}$

6.9 Roche radius: $a_{\rm Roche} = r_{\rm p}\sqrt[3]{\frac{3\,M_\star}{m_{\rm p}}}$

6.10 mass parcel: $\delta m = \frac{1}{2}\,k_2\,m_{\rm p}\left(\frac{r_{\rm p}}{a_{\rm peri}}\right)^3$

6.11 change in orbital energy: $\dot{E} = \frac{dE}{dt} = \frac{2\pi\,\Delta E}{Q\,T}$

6.12 energy to deform the planet: $\Delta E = \frac{1}{2}\,\Delta I\,\omega^2$

6.13 condition for pseudo synchronisation: $\omega = \frac{v_{\rm peri}}{a_{\rm peri}}$

6.14 Inertia for a spherical body: $I_{\rm sph} = \frac{2}{5}\,m_{\rm p}\,r_{\rm p}^2$

6.15 Inertia for a dumbbell: $I_{\rm dum} = 2\,\delta m\,r_{\rm p}^2$

6.16 Change in inertia: $\Delta I = \frac{6}{5}\,\delta m\,r_{\rm p}^2$

6.17 Change in orbital energy: $\Delta E = \frac{3}{5}\,k_2\,\left(\frac{r_{\rm p}}{a_{\rm peri}}\right)^5\,\frac{(1+e)}{(1-e)}\,E$

6.18 Tidal timescale: $\frac{1}{\tau_{\rm tides}} = \frac{\dot{E}}{E}$

6.19 Tidal timescale: $\tau_{\rm tides} = \frac{5}{6\pi}\frac{Q\,T}{k_2}\,\left(\frac{a_{\rm peri}}{r_{\rm p}}\right)^5\,\frac{1-e}{1+e} \propto \left(\frac{a_{\rm peri}}{r_{\rm p}}\right)^5$

6.20 Tidal circularisation timescale: $\tau_{\rm circ} = \frac{4}{63}Q\sqrt{\frac{a^3}{GM_\star}}\left(\frac{m_{\rm p}}{M_\star}\right)\left(\frac{a}{r_{\rm p}}\right)^5$

6.21 Tidal synchronisation timescale: $\tau_{\rm syn} = Q\left(\frac{r_{\rm p}^3}{Gm_{\rm p}}\right)\left(\frac{m_{\rm p}}{M_\star}\right)^2\left(\frac{a}{r_{\rm p}}\right)^6(\omega_{\rm p}-\omega_{\rm syn})$

6.22 Tidal infall timescale: $\tau_{\rm infall} \propto \left(\frac{a}{R_\star}\right)^5\left(\frac{M_\star}{m_{\rm p}}\right)$

6.23 Rossiter-McLaughlin effect amplitude: $K_{\rm RM} = \frac{2}{3}D\,v\sin i_\star\sqrt{1-b^2}$

6.24 Spin-orbit angle: $\cos\psi = \sin i_\star\cos\lambda\sin i_{\rm p} + \cos i_\star\cos i_{\rm p}$

Astrophysical Constants and Units

Astronomical Unit	AU	1.50×10^{11} m
Parsec	pc	3.1×10^{16} m
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_\odot$	3.84×10^{26} W
Absolute bolometric magnitude of the Sun	$M_{\rm bol}$	$+4^{\rm m}75$
Bolometric correction for the Sun	BC	$-0^{\rm m}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\rm m}74$
Solar constant	f	1.36×10^{3} W m^{-2}
Solar mass	$M_\odot$	1.989×10^{30} kg
Solar radius	$R_\odot$	6.95×10^{8} m
Solar effective temperature	$T_{\rm eff}$	5777 K
Wien constant	b	2.898×10^{-3} m K
Hubble constant	H_o	70 km s^{-1} Mpc^{-1}
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_\oplus$	5.972×10^{24} kg
Earth radius	$R_\oplus$	6371 km
Jupiter mass	$M_{\rm Jup}$	1.898×10^{27} kg
Jupiter radius	$R_{\rm Jup}$	71 492 km

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.